Andrea Daly

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Physics and astronomy Masters graduate with research and hands-on experience in CCD imaging, single slit spectroscopy, and echelle spectroscopy on both small and large aperture telescopes. Demonstrated leadership and teaching experience through mentoring undergrad research projects and teaching undergraduate courses.

Education

University of Wyoming **August 2022 – May 2024**
Master of Science in Physics, cGPA:3.14 *Laramie, WY*

Embry-Riddle Aeronautical University **August 2018 – May 2022**
Bachelor of Science in Astronomy, Minor in Computer Science, cGPA: 3.8 *Prescott, AZ*

Relevant Coursework

- Astronomical Techniques
- Stellar Evolution
- Cosmology
- Galaxies
- Planets and their Stars
- Computer Science I & II
- Techniques of Observational Astronomy
- Organization of Programming Languages

Experience

Graduate Assistant **August 2022 – May 2024**
University of Wyoming
- Fully certified to virtually observe on Apache Point Observatory ARC Echelle Spectrograph (ARCES) using the Telescope User Interface (TUI)
- Fit multi-parameter models to spectral energy distributions
- As a teachers assistant, led weekly labs which included the operating and setup of lab equipment, grading, and one-on-one instruction of materials, resulting in a cumulative course evaluation score of 4.5/5

Research Assistant **August 2019 – May 2022**
Embry-Riddle Aeronautical University
- Trained to use the on-campus telescope and other equipment
- Manipulation and synthesis of fits files from various observatories and catalogs, resulting in publications

Projects

Prescott Observatory Team for Analyzing Telescopically Observed Spectra (POTATOS) | *ERAU* **2021 – 2022**
- Recruited and trained freshman and sophomores to become members of the team and operate the on-campus telescope and reduce the echelle data
- Observed O-star binary systems,to constrain stellar parameters of radii, temperature, and masses

The long-period orbit of the dust-producing Wolf-Rayet binary WR 125 | *ERAU* **2020 – 2022**
- Used spectra from Keck Observatory and Gemini Observatory, measured the radial velocities for the system to constrain the orbit
- Analyzed the colliding winds of the system using SOFIA/FORCAST grism spectroscopy

How does dust form near Wolf-Rayet stars? | *ERAU* **2020**
- Learned the observation and data reduction process using the on-campus telescope and other equipment
- Analysis of colliding wind binary system WR 125

Spectroscopy of Be Stars Observed with TESS | *ERAU* **2019**
- Focused on massive binaries and colliding winds of binary star systems
- Analysis and manipulation of fits files

Skills

Programming Languages: Python, Java, C, C++, JavaScript, MATLAB, IRAF
Trained in: LaTeX, Microsoft Office, American Sign Language, MIG Welding

Leadership and Conferences

- 'Two in a million - The interplay between binaries and star clusters' Conference at European Southern Observatory (2024)
- Vice President of Sigma Pi Sigma at ERAU (2021 – 2022)
- Outreach Coordinator for Ambassadors Program at ERAU (2021)
- Conference for Undergraduate Women in Physics (2019, 2020)

Grants and Awards

Graduate Education Grant **2022 – 2024**
$32,290 *University of Wyoming*

Wyoming NASA Space Grant **2023**
$20,000 *University of Wyoming*

URI Ignite Award and Arizona Space Grant **2019 – 2022**
$6,000 *Undergraduate Research Institute ERAU*

Excellence Award **2018 – 2022**
$5,000 *ERAU*

Presidential Scholarship **2018 – 2022**
$16,000 *ERAU*

Publications

- Richardson, N., **Daly, A.**, N., Chené, A., Hill, G., Williams, P., Shenavrin, V., Weigelt, G. (2024). The Long Period Orbit of the Dust-Producing Wolf-Rayet Binary WR 125. (arXiv:2405.10454)

- Richardson, N. D., Pablo, H., Barclay, K., Beltran, M., **Daly, A.**, Elliott, A., Lane, A., Medina, E., Pavao, C., Peatt, M., Perez, L., Pickett, C., Williams, A., & Rachford, B. (2021). Binary Parameters for the Massive Eclipsing Binary CC Cassiopeiae. Paper presented at the 110.10.5281/zenodo.5129293 https://ui.adsabs.harvard.edu/abs/2021tsc2.confE.110R

- Richardson, N. D., Thizy, O., Bjorkman, J. E., Carciofi, A., Rubio, A. C., Thomas, J. D., Bjorkman, K. S., Labadie-Bartz, J., Genaro, M., Wisniewski, J. P., Wang, L., Gies, D. R., Chojnowski, S. D., **Daly, A.**, Edwards, T., Fowler, C., Gullingsrud, A. D., Habel, N., James, D. J., . . . Waldschlaeger, U. (2021). Outbursts and stellar properties of the classical Be star HD 6226. Monthly Notices of the Royal Astronomical Society, 508, 2002-2018. 10.1093/mnras/stab2759; eprintid: arXiv:2109.11026

- Fichou, A., Kitsu, B., Thigpen, M., Wahlstrom, B., Negrete, A., Wong, S., March, R., Warren, C., Sherman, C., McSheehy, E., Rodriguez, C., Bowers, C., Roberts, J., Manacas, R., Morando-Hernandez, N., Iordanov, A., Haye, E., Brand, Z., **Daly, A.**, Fussell, G. C., Pledger, J., Rossi, N., Molitor, R., & LaBate, E. (2020). E.R.E.B.U.S (PROTOTYPE). , (2020). https://commons.erau.edu/pr-undergraduate-works/3